ELEC 231 — Mathematical Methods I for ECE · Fall 2026

# Pre-tutorial 1

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**Note (Instructions).** Work through the problem before your tutorial and upload your solution to onQ as a single PDF by 9:00 AM on the due date (no in-class submission). Show your working, not just the final answer: a genuine, complete attempt earns the 2 marks for this pre-tutorial; a missing or incomplete attempt earns 0.

## Problem

*Context.* Week 1 notes (Nilsson & Riedel, "Natural response of first-order RL and RC circuits"): the circuit of Fig. T1.1 has been in place with the switch closed for a long time; the switch opens at $t = 0$. For $t \geq 0$ the source and $R_0$ are disconnected and only the loop of Fig. T1.2 remains.

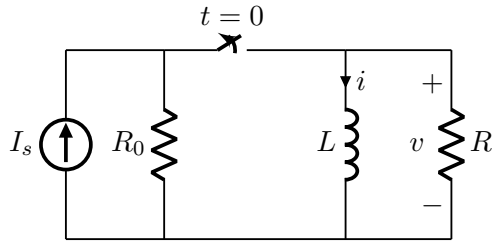



**Figure T1.1.** Fig. 1 of the Week 1 notes: switch closed for a long time, opens at $t = 0$.

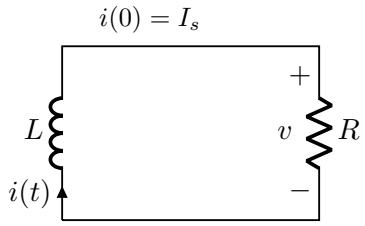



**Figure T1.2.** Fig. 2 of the Week 1 notes: the circuit for $t \geq 0$, with $i(0) = I_s$.

**Problem 1 (Tutorial problem, Week 1 notes p. 8).** If $L = 8\,\text{mH}$ and $R = 4\,\Omega$ in Fig. 2, and $I_s = 4\,\text{A}$, what is the numerical expression for $i(t)$, $t \geq 0$? Sketch $i(t)$, $t \geq 0$.

Source: Prof. Korenberg, Week 1 handwritten notes, last page, heading "Tutorial Problem" (posted 6 Sep 2025).